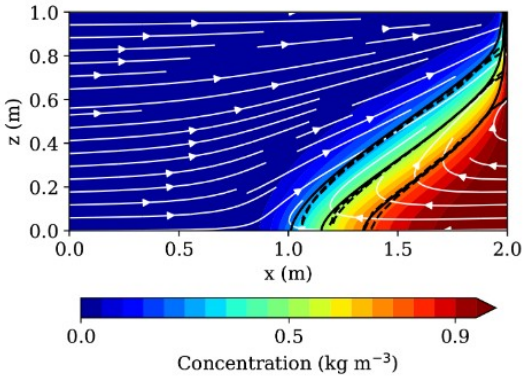
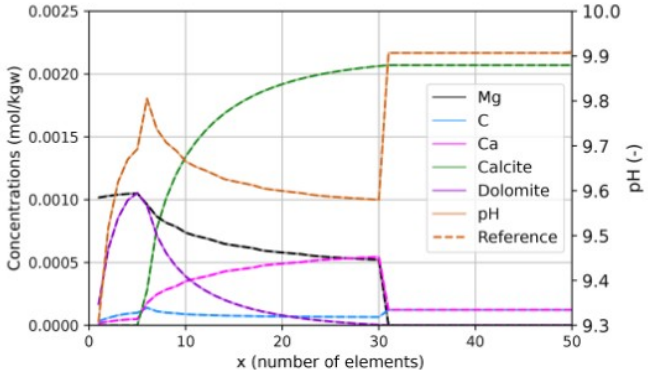
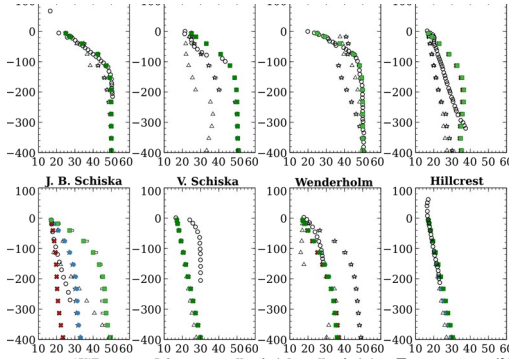
TRANSPORT Simulation Environment

Installation

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Software prerequisites: uv and git

uv is the Python package and project manager used by TRANSPORTSE. It replaces mamba/conda.

```
$ curl -LsSf https://astral.sh/uv/install.sh | sh # Linux, macOS
PS> powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
$ pipx install uv # alternatives: pip install uv | brew install uv
```

Documentation: https://docs.astral.sh/uv | source: https://github.com/astral-sh/uv

git (version control):

```
$ sudo apt install git # Debian/Ubuntu
$ brew install git # macOS
```

Check the installation: uv --version | git --version

Clone the TRANSPORTSE git repository

GFZ GitLab (primary repository):

```
$ git clone https://git.gfz-potsdam.de/kempka/transportse.git
$ cd transportse
```

Alternative - public GitHub mirror:

```
$ git clone https://github.com/thkempka/transportse.git
```

pyproject.toml and uv.lock in the repository provide the complete, pinned dependency set.

Install the Python environment

uv resolves the pinned dependency set from pyproject.toml/uv.lock and installs it into a local .venv - an environment.yml is no longer required.

```
$ uv sync
```

uv init is NOT needed for TRANSPORTSE: it only scaffolds a new project (pyproject.toml with an empty dependency list) and does not add any requirements.

Reproducible install from the lock file: uv sync --frozen

uv add <package> is the command that writes new requirements into pyproject.toml and updates uv.lock.

Alternative: add individual libraries to your own project

To use the TRANSPORTSE libraries in an existing uv project, add the relevant packages to that project:

```
$ uv add joblib numba numpy phreeqpy pyevtk ipython
```

uv add writes the requirements into pyproject.toml and updates uv.lock - the equivalent of mamba install plus a manual environment.yml edit.

Development tools (e.g. pytest):

```
$ uv add --dev pytest
```

Run TRANSPORTSE

```
$ uv run python tse.py config.py
```

uv run executes the command inside the project environment - no manual activation required.

Alternatively activate the environment:

```
$ source .venv/bin/activate # Windows: .venv\Scripts\activate
$ python tse.py config.py
```